Road Safety and Accident Severity in the United States: A Case Study of Texas

ECON 5140: Applied Econometrics

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Motivation and Research Question

Which weather and roadway factors are associated with higher crash severity in Texas?

Traffic accidents create major medical, economic, and productivity costs.

Reducing severity is as important as reducing crash frequency.

Weather and road design may significantly affect outcomes.

These findings can inform policy, infrastructure, and emergency response.

❑ Identification

Is This Causal?

- No direct causal claim is made.
- Weather and nighttime conditions are not randomly assigned.
- Results are interpreted as conditional associations after controlling for observed factors.

✓ Several models are used to evaluate severity outcomes in different ways. Comparing results across models helps assess reliability.

❑ Overview

- **Data**

U.S. Accidents Dataset (2016–2023)

Texas case study

Original sample: 572,795 observations

Working sample: 20,000 observations

- **Plan:**

1. OLS regression benchmark
2. Ordered Logit main model
3. Binary Logit for severe crash probability
4. Marginal effects
5. Robustness checks
6. Clustered standard errors
7. Model evaluation

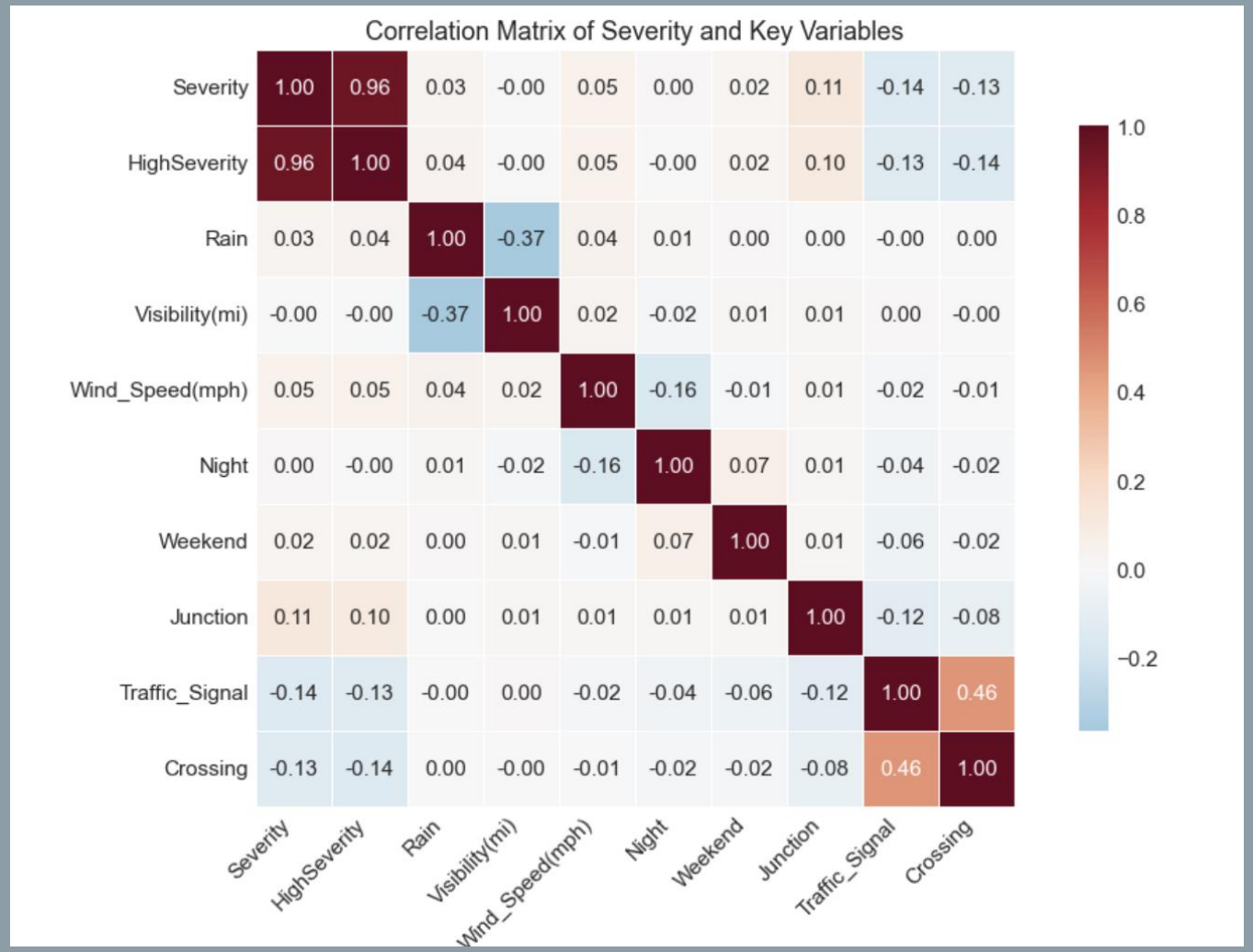
Outcomes
Severity (1–4)
HighSeverity = 1 if Severity ≥ 3
Severe4 = 1 if Severity = 4

Explanatory Variables

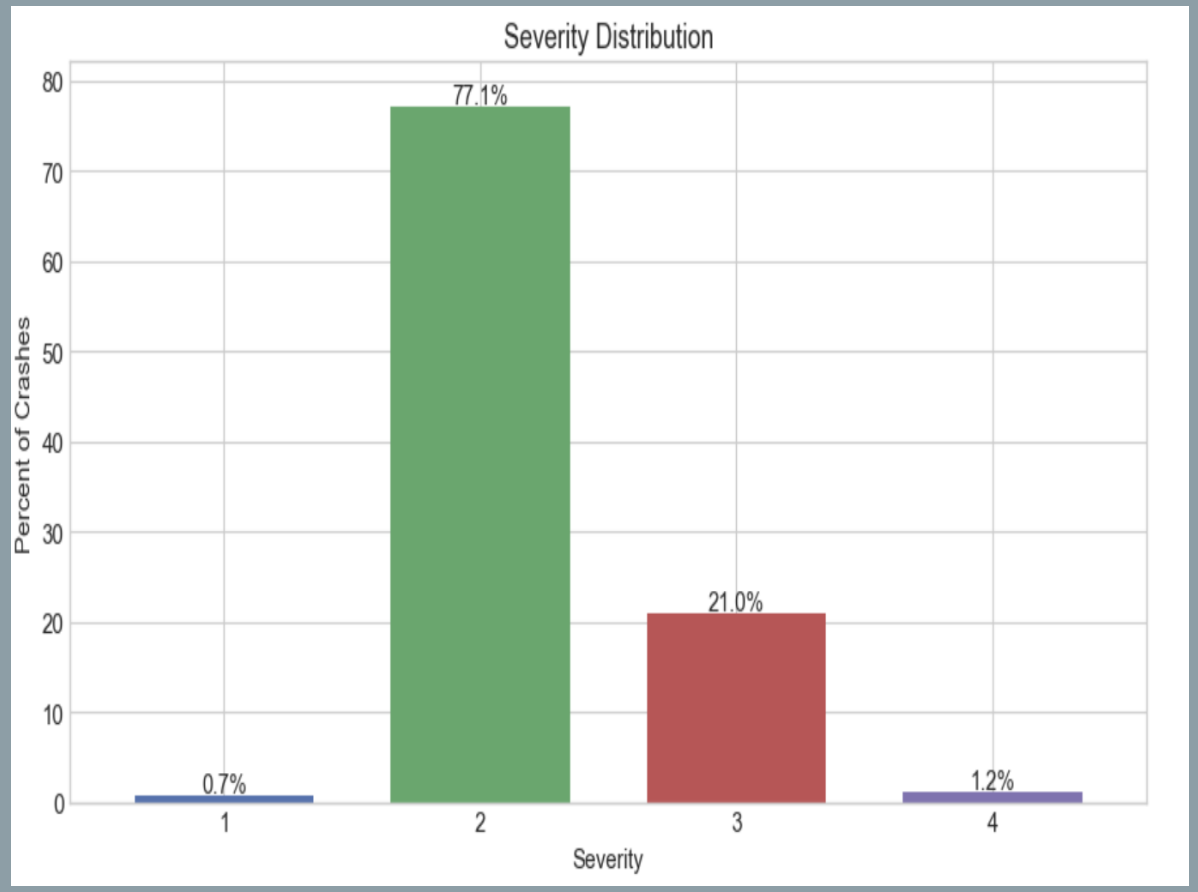
- Rain, Visibility, Temperature, Wind Speed, Night, Weekend, Rush Hour, Junction, Traffic Signal, Crossing

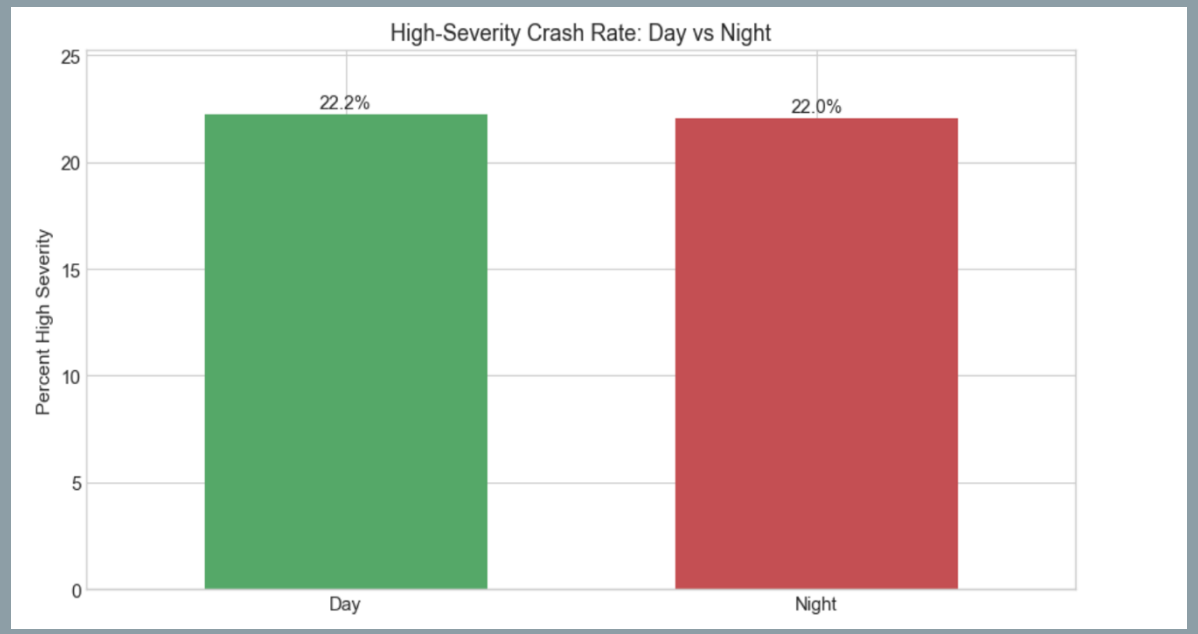
❑ Sample Statistics and Correlation

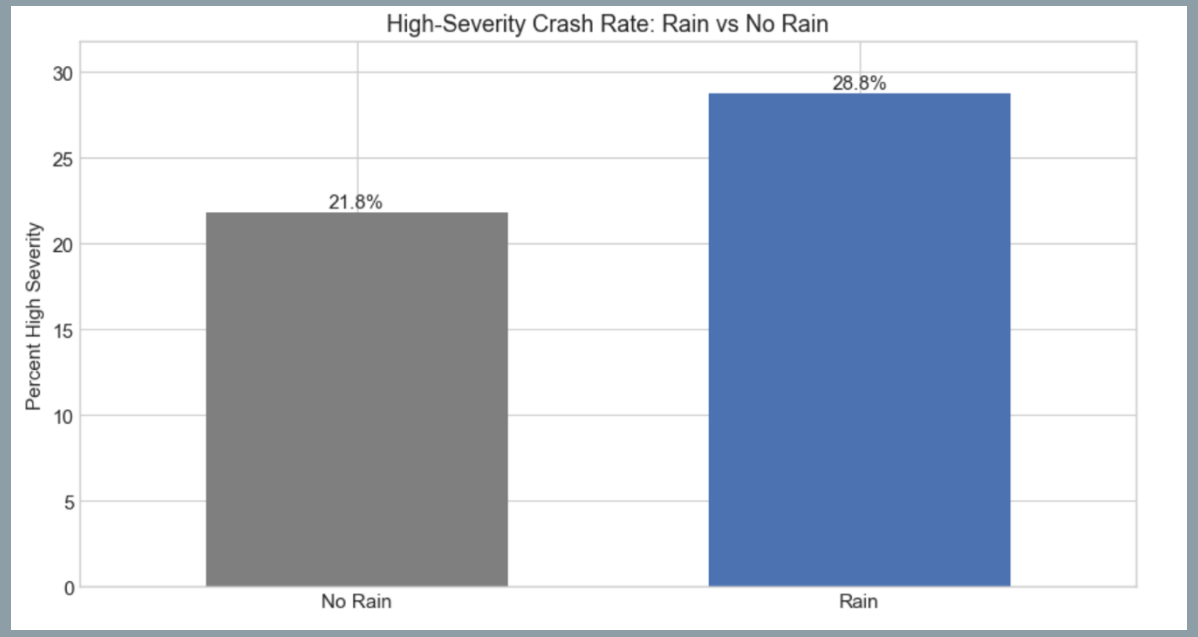
	mean	std	min	max
Severity	2.227	0.461	1.000	4.000
HighSeverity	0.222	0.415	0.000	1.000
Severe4	0.012	0.108	0.000	1.000
Visibility(mi)	9.240	2.151	0.000	25.000
Precipitation(in)	0.005	0.045	0.000	1.410
Temperature(F)	70.110	16.359	1.000	109.000
Wind_Speed(mph)	8.585	4.854	0.000	41.000
Rain	0.051	0.219	0.000	1.000
Night	0.280	0.449	0.000	1.000
Weekend	0.131	0.337	0.000	1.000
RushHour	0.414	0.493	0.000	1.000



❑ Key Patterns







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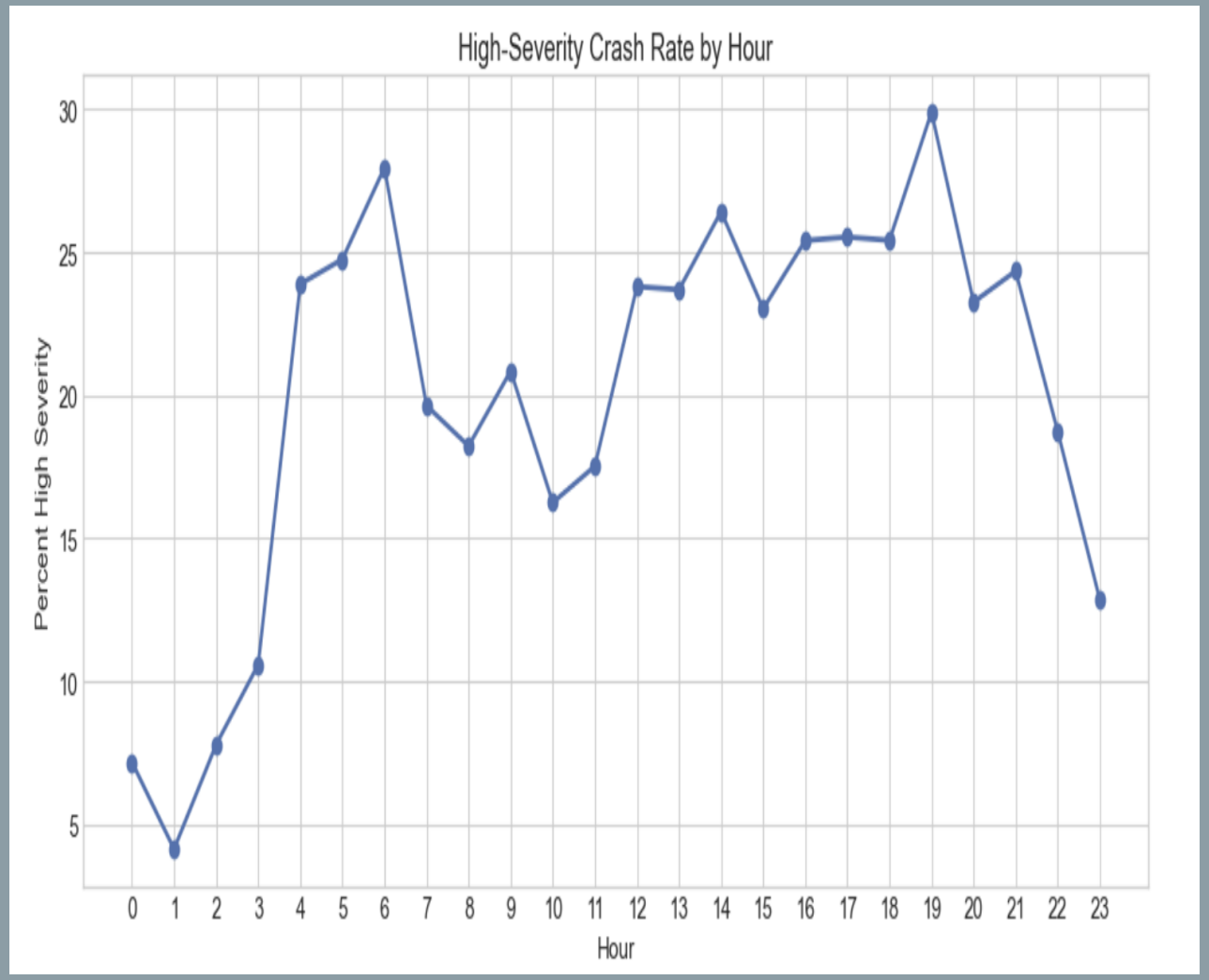

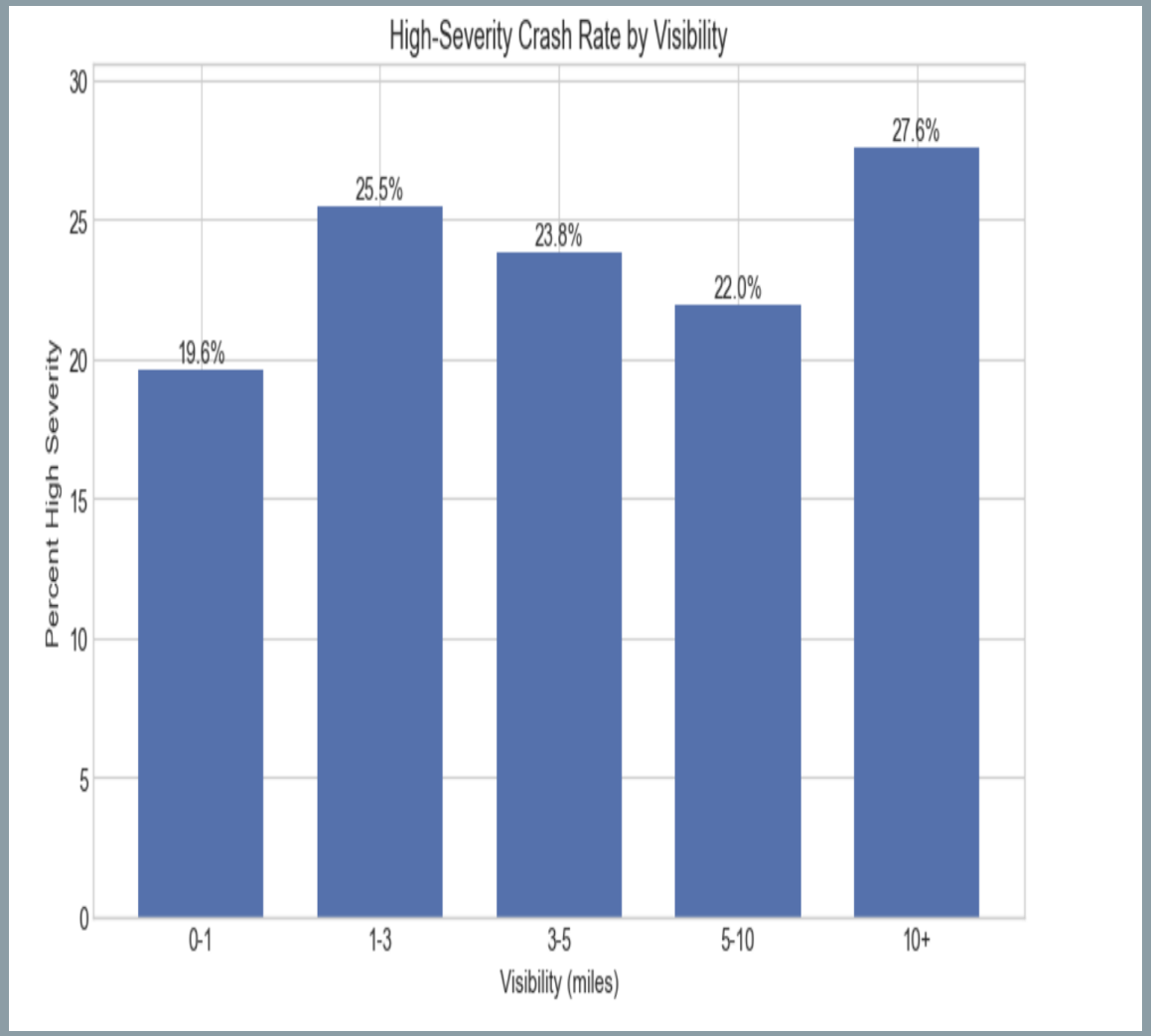

❑ OLS Benchmark Model

```
                            OLS Regression Results
==============================================================================
Dep. Variable:               Severity   R-squared:                       0.088
Model:                            OLS   Adj. R-squared:                  0.086
Method:                 Least Squares   F-statistic:                     89.17
Date:                Fri, 17 Apr 2026   Prob (F-statistic):               0.00
Time:                        19:50:53   Log-Likelihood:                -11984.
No. Observations:               20000   AIC:                         2.404e+04
Df Residuals:                   19963   BIC:                         2.434e+04
Df Model:                          36
Covariance Type:                  HC1
==========================================================================================
                                     coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------
Intercept                          2.2158      0.028     79.577      0.000       2.161       2.270
Junction[T.True]                   0.1482      0.018      8.192      0.000       0.113       0.184
Traffic_Signal[T.True]            -0.0625      0.007     -8.689      0.000      -0.077      -0.048
Crossing[T.True]                  -0.1245      0.008    -15.655      0.000      -0.140      -0.109
C(County_Grouped)[T.Collin]       -0.1484      0.029     -5.065      0.000      -0.206      -0.091
C(County_Grouped)[T.Dallas]        0.0337      0.015      2.236      0.025       0.004       0.063
C(County_Grouped)[T.Denton]        0.2235      0.040      5.627      0.000       0.146       0.301
C(County_Grouped)[T.El Paso]      -0.2306      0.021    -10.811      0.000      -0.272      -0.189
C(County_Grouped)[T.Fort Bend]    -0.0549      0.058     -0.948      0.343      -0.168       0.059
C(County_Grouped)[T.Harris]       -0.0610      0.014     -4.234      0.000      -0.089      -0.033
C(County_Grouped)[T.Hays]          0.2112      0.074      2.865      0.004       0.067       0.356
C(County_Grouped)[T.Montgomery]    0.2902      0.063      4.604      0.000       0.167       0.414
C(County_Grouped)[T.Other]         0.2449      0.033      7.483      0.000       0.181       0.309
C(County_Grouped)[T.Rockwall]      0.3737      0.091      4.129      0.000       0.196       0.551
C(County_Grouped)[T.Smith]        -0.2706      0.016    -17.351      0.000      -0.301      -0.240
C(County_Grouped)[T.Tarrant]       0.0979      0.021      4.709      0.000       0.057       0.139
C(County_Grouped)[T.Travis]       -0.1450      0.015     -9.981      0.000      -0.173      -0.117
C(County_Grouped)[T.Wichita]      -0.2732      0.014    -18.848      0.000      -0.302      -0.245
C(County_Grouped)[T.Williamson]   -0.0819      0.034     -2.413      0.016      -0.148      -0.015
C(Month)[T.2]                      0.0130      0.014      0.910      0.363      -0.015       0.041
C(Month)[T.3]                      0.0391      0.015      2.558      0.011       0.009       0.069
C(Month)[T.4]                      0.0332      0.016      2.052      0.040       0.001       0.065
C(Month)[T.5]                      0.0424      0.018      2.387      0.017       0.008       0.077
C(Month)[T.6]                      0.0352      0.019      1.896      0.058      -0.001       0.072
C(Month)[T.7]                      0.0053      0.018      0.291      0.771      -0.030       0.041
C(Month)[T.8]                      0.0176      0.017      1.016      0.310      -0.016       0.052
C(Month)[T.9]                      0.0193      0.016      1.171      0.241      -0.013       0.052
C(Month)[T.10]                     0.0344      0.015      2.274      0.023       0.005       0.064
C(Month)[T.11]                    -0.0147      0.013     -1.110      0.267      -0.041       0.011
C(Month)[T.12]                    -0.0115      0.013     -0.869      0.385      -0.037       0.014
Q("Visibility(mi)")            -1.855e-05      0.002     -0.012      0.991      -0.003       0.003
Rain                               0.0556      0.016      3.518      0.000       0.025       0.087
Q("Temperature(F)")                0.0004      0.000      1.368      0.171      -0.000       0.001
Q("Wind_Speed(mph)")               0.0020      0.001      2.899      0.004       0.001       0.003
Night                              0.0018      0.008      0.228      0.820      -0.014       0.018
RushHour                          -0.0035      0.007     -0.523      0.601      -0.017       0.010
Weekend                            0.0144      0.010      1.508      0.132      -0.004       0.033
==============================================================================
Omnibus:                     3810.335   Durbin-Watson:                   1.995
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             6728.875
Skew:                           1.228   Prob(JB):                         0.00
Kurtosis:                       4.428   Cond. No.                     1.56e+03
==============================================================================
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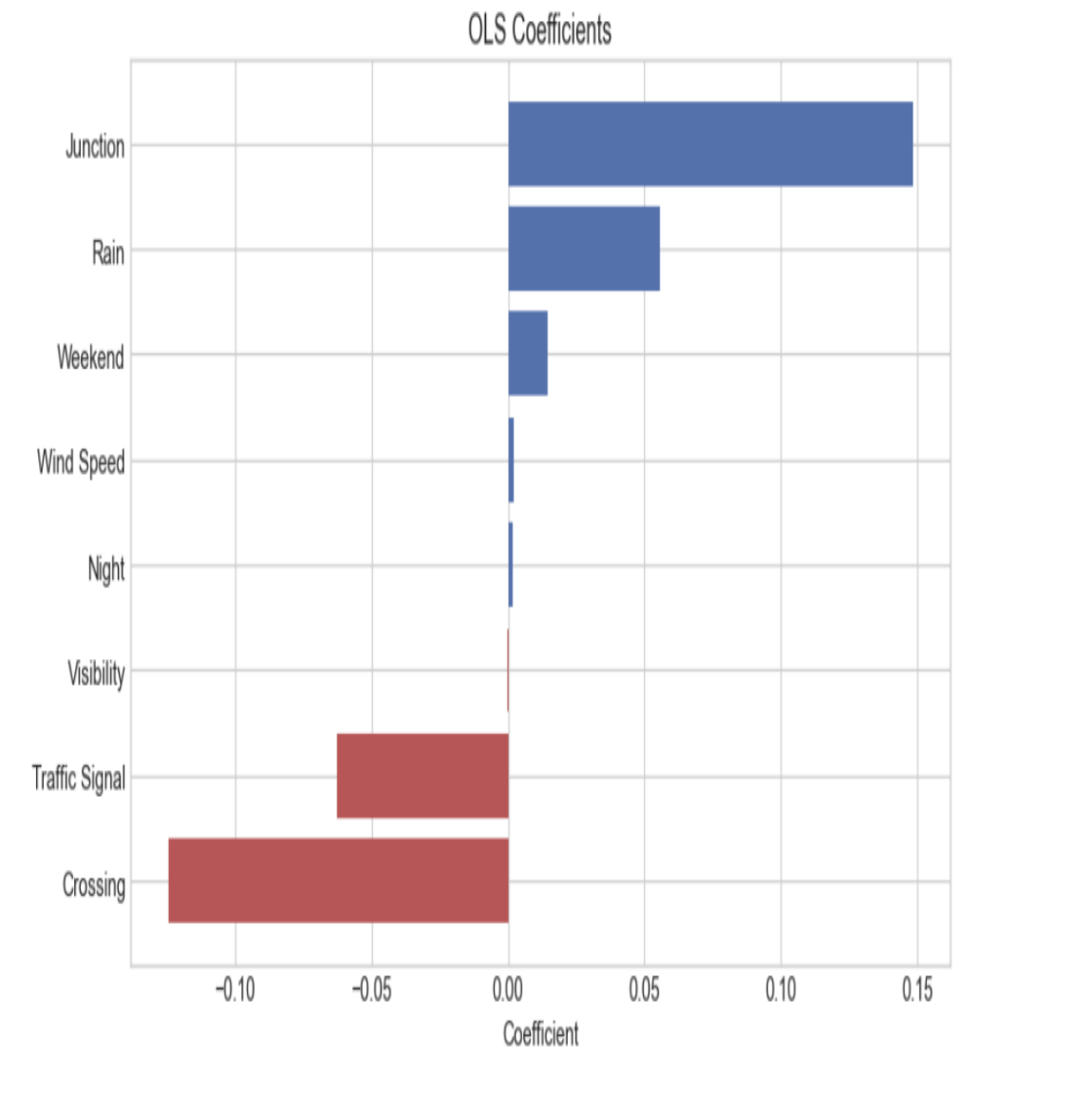
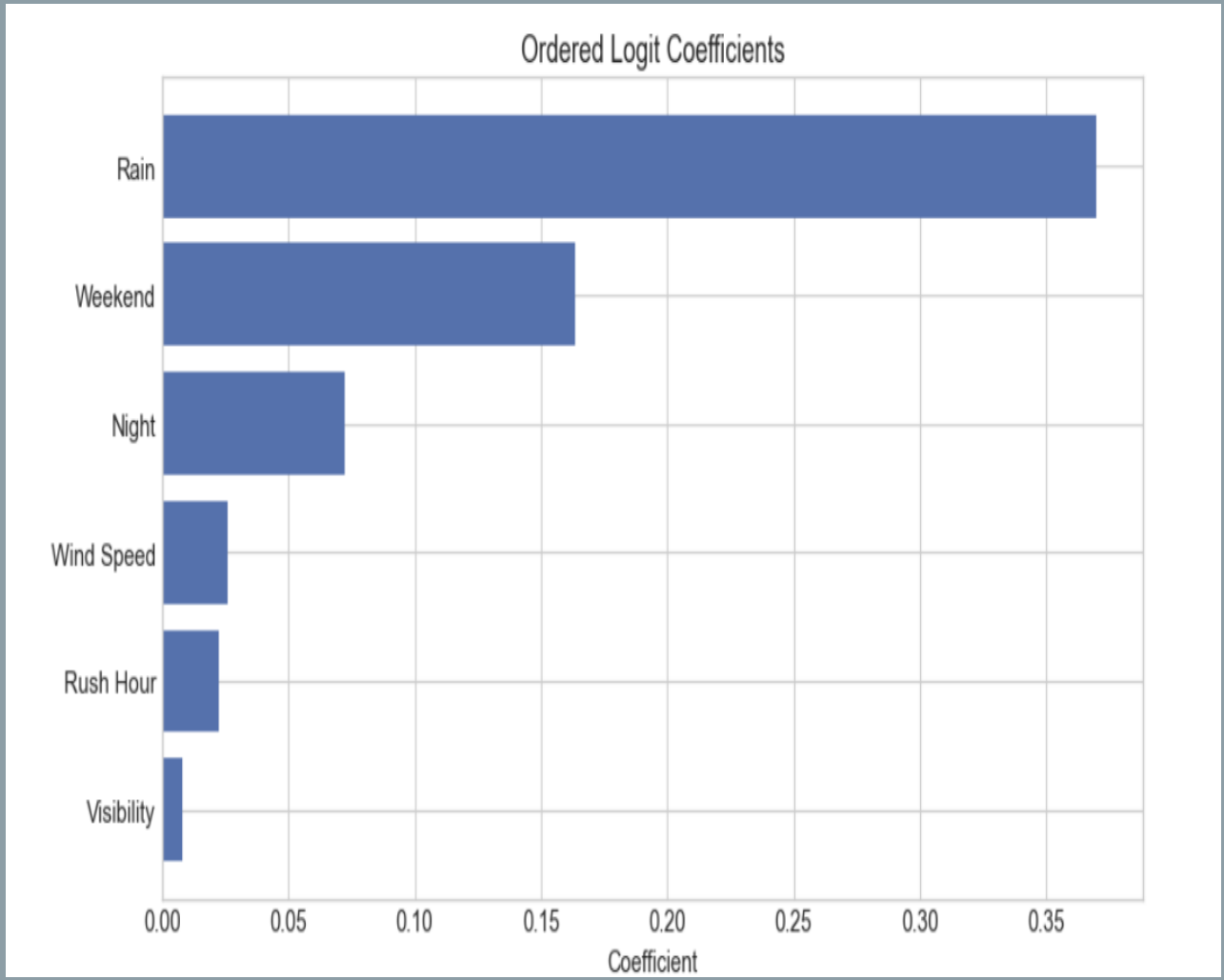


❑ Ordered Logit Model

```
                           OrderedModel Results
==============================================================================
Dep. Variable:               Severity   Log-Likelihood:               -12253.
Model:                   OrderedModel   AIC:                        2.453e+04
Method:            Maximum Likelihood   BIC:                        2.461e+04
Date:                 Fri, 17 Apr 2026
Time:                         19:52:30
No. Observations:               20000
Df Residuals:                   19990
Df Model:                           7
==================================================================================
                     coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------
Visibility(mi)     0.0080      0.009      0.925      0.355      -0.009       0.025
Rain               0.3697      0.078      4.766      0.000       0.218       0.522
Temperature(F)     0.0025      0.001      2.222      0.026       0.000       0.005
Wind_Speed(mph)    0.0262      0.003      7.547      0.000       0.019       0.033
Night              0.0721      0.042      1.698      0.089      -0.011       0.155
RushHour           0.0222      0.037      0.607      0.544      -0.050       0.094
Weekend            0.1637      0.049      3.356      0.001       0.068       0.259
1/2               -4.4272      0.142    -31.215      0.000      -4.705      -4.149
2/3                1.8294      0.014    132.768      0.000       1.802       1.856
3/4                1.1578      0.020     57.252      0.000       1.118       1.197
==================================================================================
```



❑ Binary Logit Model

```
                           Logit Regression Results
==============================================================================
Dep. Variable:           HighSeverity   No. Observations:                20000
Model:                          Logit   Df Residuals:                    19978
Method:                           MLE   Df Model:                           21
Date:                Fri, 17 Apr 2026   Pseudo R-squ.:                 0.04213
Time:                        19:53:54   Log-Likelihood:                -10138.
converged:                       True   LL-Null:                       -10584.
Covariance Type:            nonrobust   LLR p-value:                3.029e-175
==============================================================================================
                             coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------------------
Intercept                 -1.4361      0.140    -10.270      0.000      -1.710      -1.162
Junction[T.True]           0.7113      0.065     10.862      0.000       0.583       0.840
Traffic_Signal[T.True]    -0.4921      0.049     -9.961      0.000      -0.589      -0.395
Crossing[T.True]          -1.1403      0.087    -13.077      0.000      -1.311      -0.969
C(Month)[T.2]              0.0703      0.086      0.820      0.412      -0.098       0.238
C(Month)[T.3]              0.2721      0.088      3.081      0.002       0.099       0.445
C(Month)[T.4]              0.2799      0.089      3.139      0.002       0.105       0.455
C(Month)[T.5]              0.4279      0.095      4.526      0.000       0.243       0.613
C(Month)[T.6]              0.4243      0.100      4.242      0.000       0.228       0.620
C(Month)[T.7]              0.3284      0.102      3.212      0.001       0.128       0.529
C(Month)[T.8]              0.3519      0.099      3.557      0.000       0.158       0.546
C(Month)[T.9]              0.2622      0.095      2.747      0.006       0.075       0.449
C(Month)[T.10]             0.2039      0.088      2.321      0.020       0.032       0.376
C(Month)[T.11]            -0.0916      0.084     -1.089      0.276      -0.256       0.073
C(Month)[T.12]            -0.1421      0.083     -1.709      0.087      -0.305       0.021
Q("Visibility(mi)")        0.0073      0.009      0.810      0.418      -0.010       0.025
Rain                       0.3633      0.081      4.506      0.000       0.205       0.521
Q("Temperature(F)")       -0.0031      0.002     -1.824      0.068      -0.006       0.000
Q("Wind_Speed(mph)")       0.0253      0.004      6.877      0.000       0.018       0.032
Night                      0.0620      0.044      1.395      0.163      -0.025       0.149
RushHour                   0.0417      0.038      1.096      0.273      -0.033       0.116
Weekend                    0.1240      0.051      2.453      0.014       0.025       0.223
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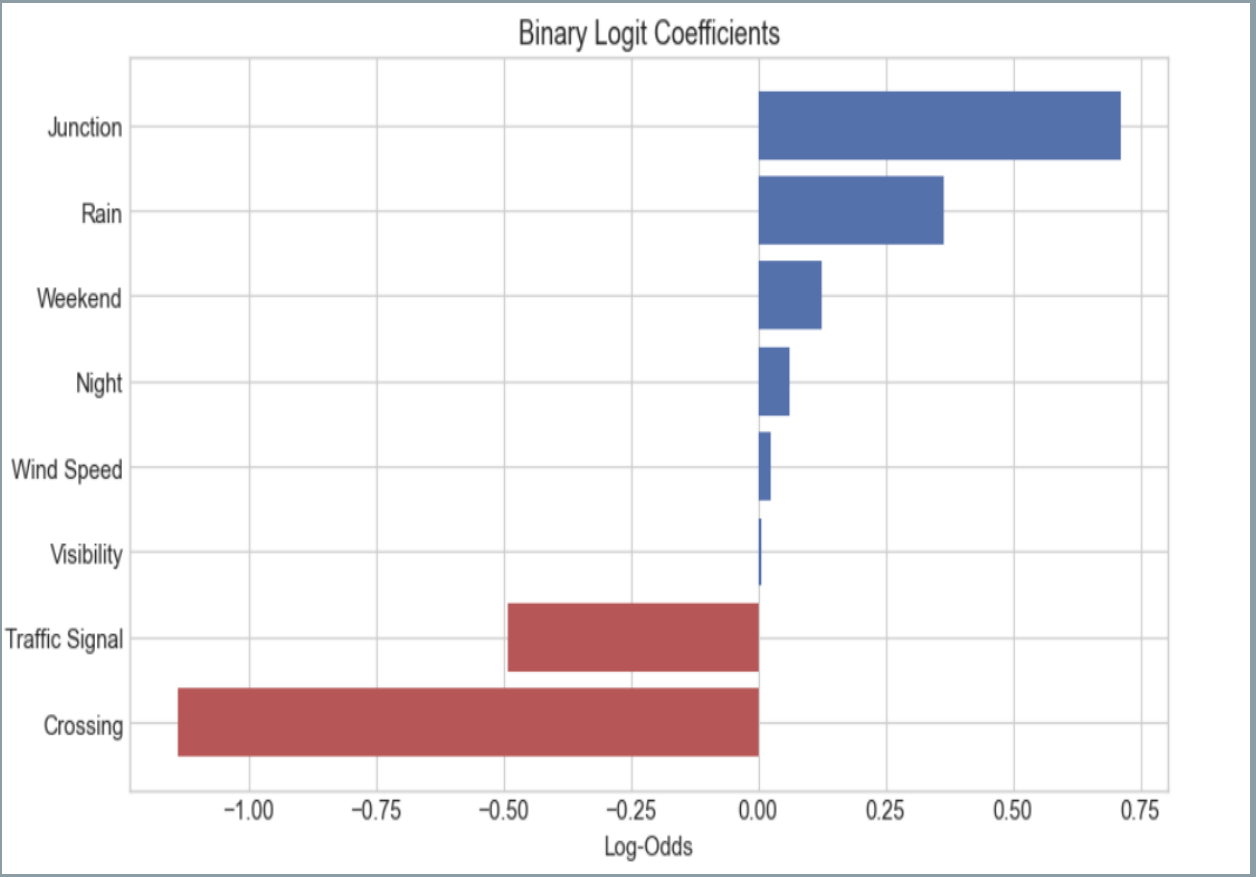
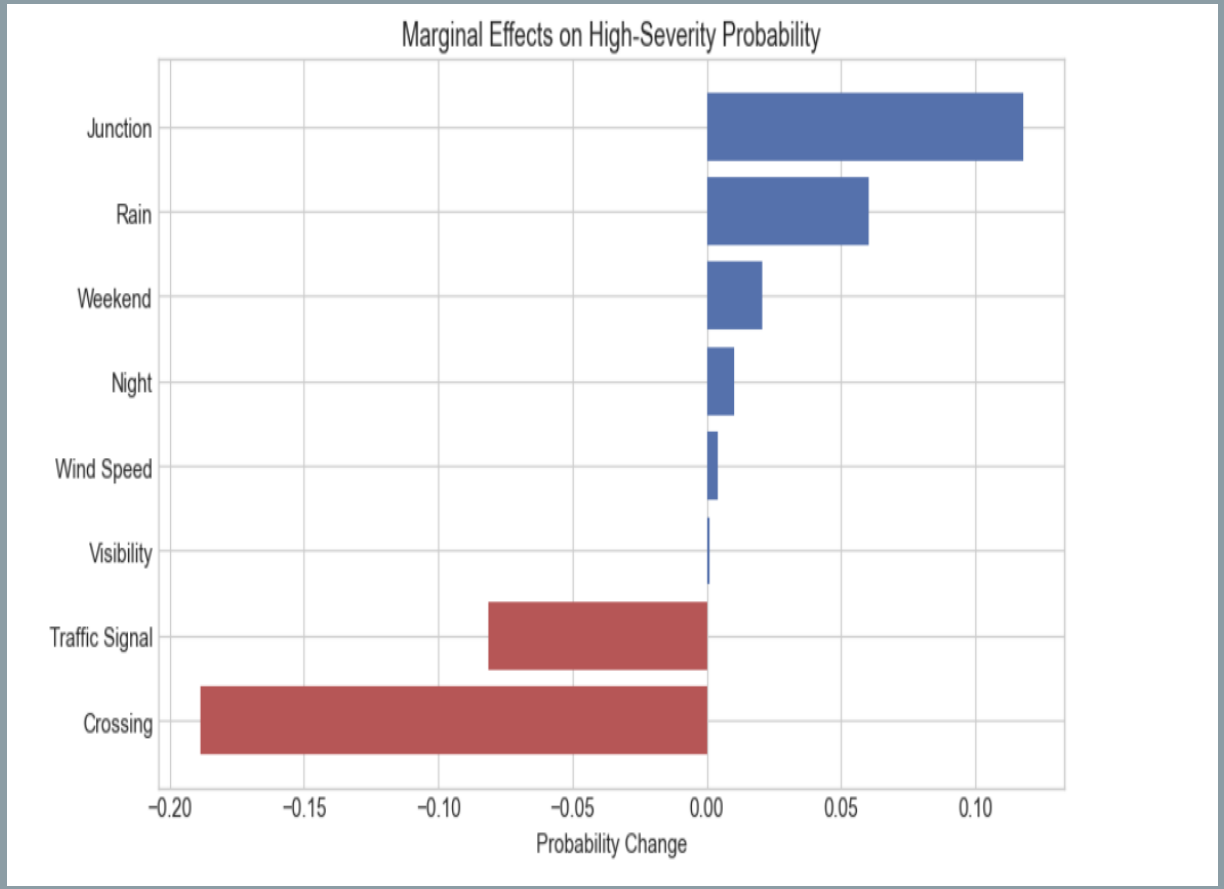


❑ Marginal Effects

Logit Marginal Effects

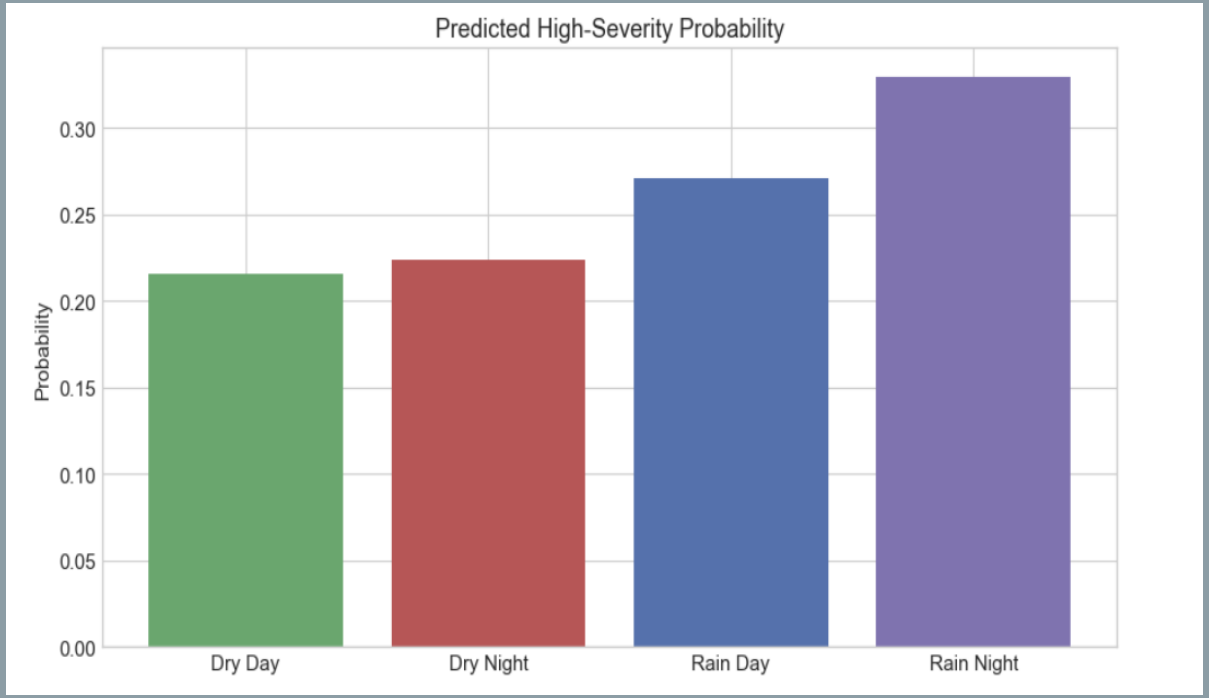
Dep. Variable: HighSeverity
Method: dydx
At: overall

	dy/dx	std err	z	P>|z|	[0.025	0.975]
Junction[T.True]	0.1177	0.011	10.976	0.000	0.097	0.139
Traffic_Signal[T.True]	-0.0814	0.008	-10.014	0.000	-0.097	-0.065
Crossing[T.True]	-0.1887	0.014	-13.132	0.000	-0.217	-0.161
C(Month)[T.2]	0.0116	0.014	0.820	0.412	-0.016	0.039
C(Month)[T.3]	0.0450	0.015	3.083	0.002	0.016	0.074
C(Month)[T.4]	0.0463	0.015	3.141	0.002	0.017	0.075
C(Month)[T.5]	0.0708	0.016	4.532	0.000	0.040	0.101
C(Month)[T.6]	0.0702	0.017	4.247	0.000	0.038	0.103
C(Month)[T.7]	0.0543	0.017	3.214	0.001	0.021	0.087
C(Month)[T.8]	0.0582	0.016	3.560	0.000	0.026	0.090
C(Month)[T.9]	0.0434	0.016	2.748	0.006	0.012	0.074
C(Month)[T.10]	0.0337	0.015	2.322	0.020	0.005	0.062
C(Month)[T.11]	-0.0152	0.014	-1.089	0.276	-0.042	0.012
C(Month)[T.12]	-0.0235	0.014	-1.709	0.087	-0.050	0.003
Q("Visibility(mi)")	0.0012	0.001	0.810	0.418	-0.002	0.004
Rain	0.0601	0.013	4.513	0.000	0.034	0.086
Q("Temperature(F)")	-0.0005	0.000	-1.824	0.068	-0.001	3.78e-05
Q("Wind_Speed(mph)")	0.0042	0.001	6.900	0.000	0.003	0.005
Night	0.0103	0.007	1.395	0.163	-0.004	0.025
RushHour	0.0069	0.006	1.096	0.273	-0.005	0.019
Weekend	0.0205	0.008	2.454	0.014	0.004	0.037

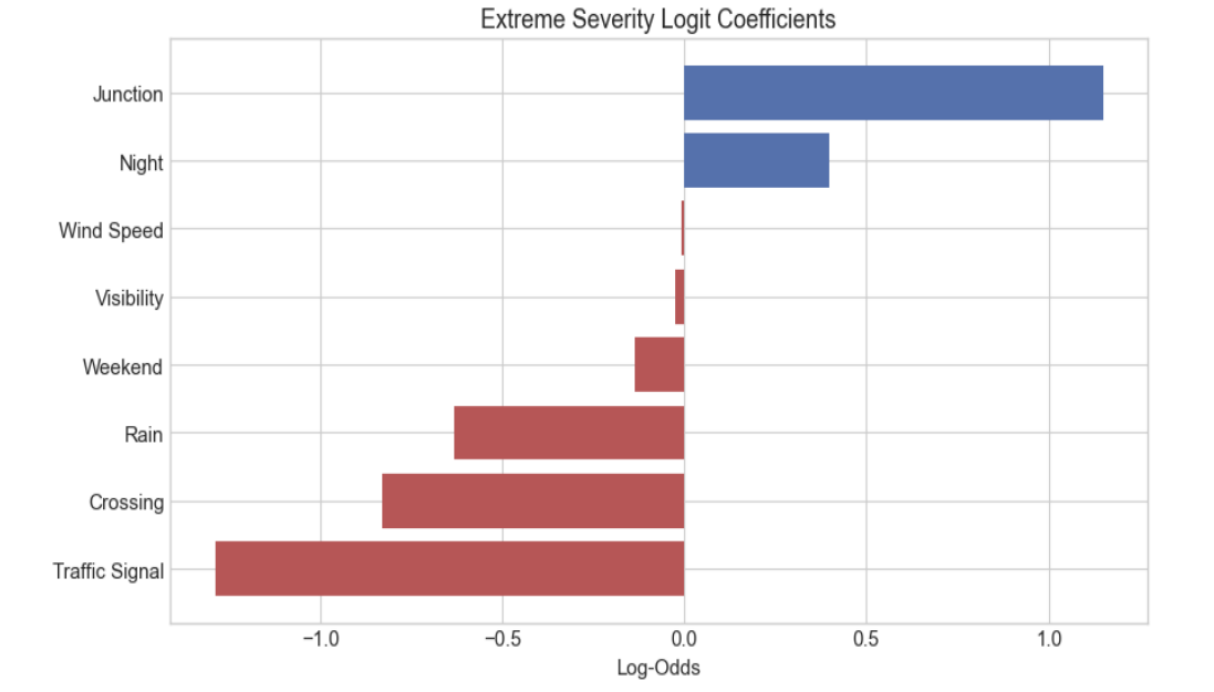


❑ Robustness Checks

Rain and Night Interaction



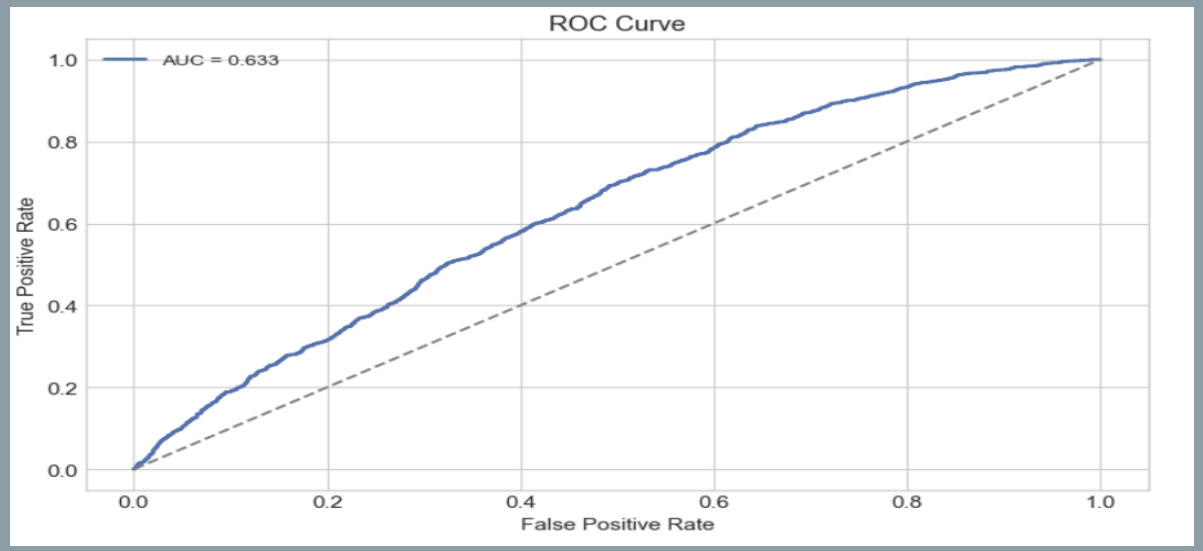
Extreme Severity Outcome

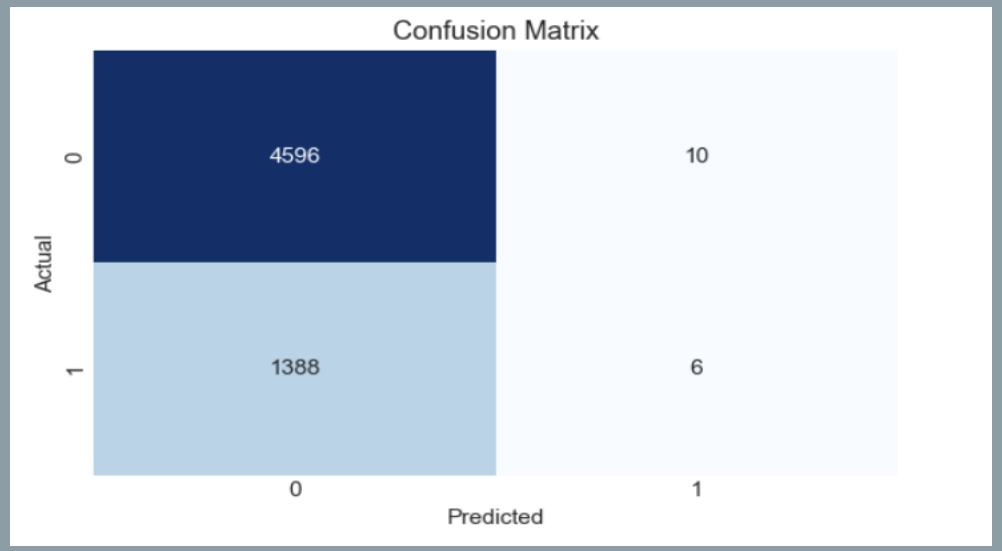


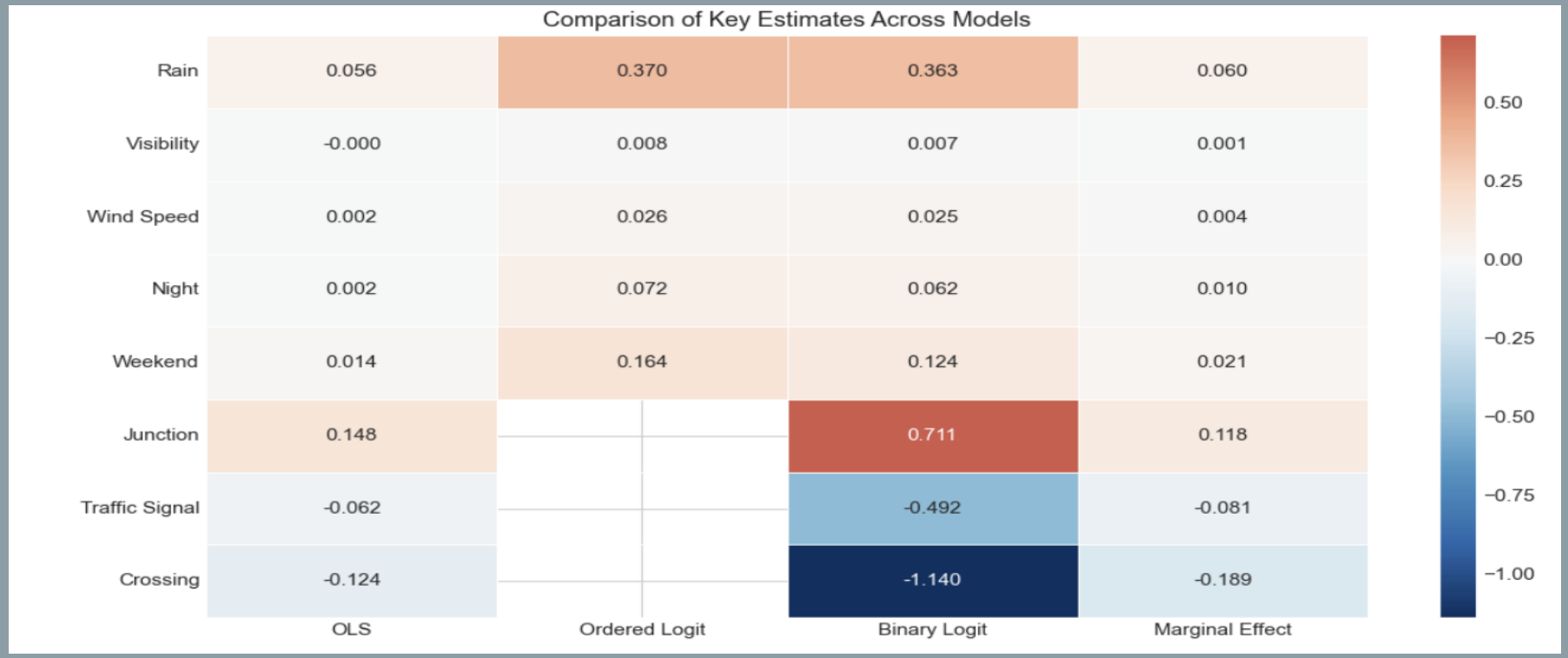
❑ County-Clustered Standard Errors

	Coefficient	SE Robust	SE Clustered
Rain	0.363300	0.080600	0.155400
Wind Speed	0.025300	0.003700	0.009900
Junction	0.711300	0.065500	0.248700
Traffic Signal	-0.492100	0.049400	0.054100
Crossing	-1.140300	0.087200	0.224300

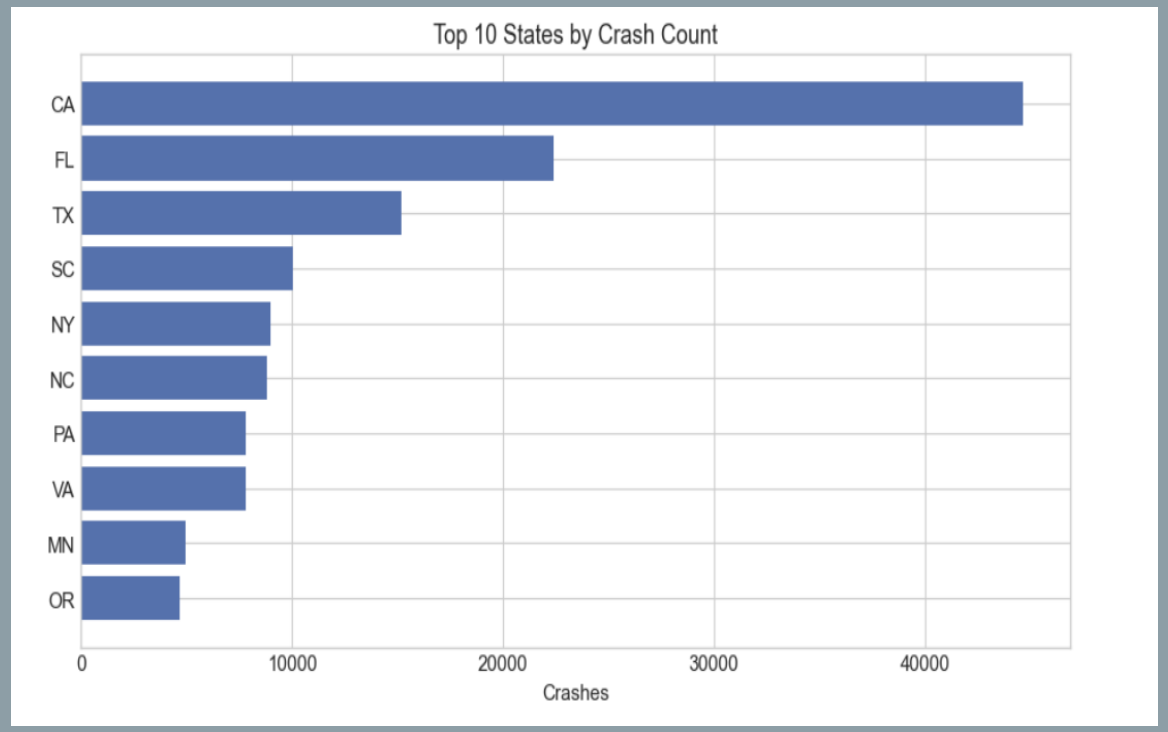
Model Evaluation

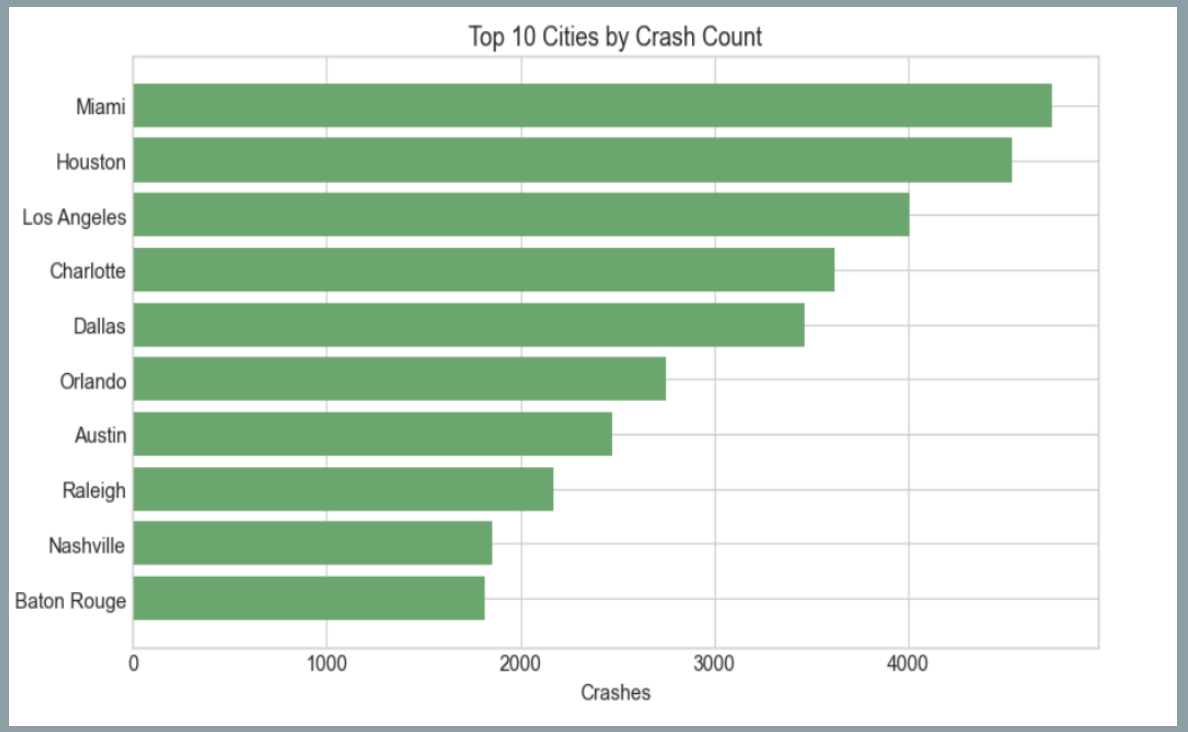


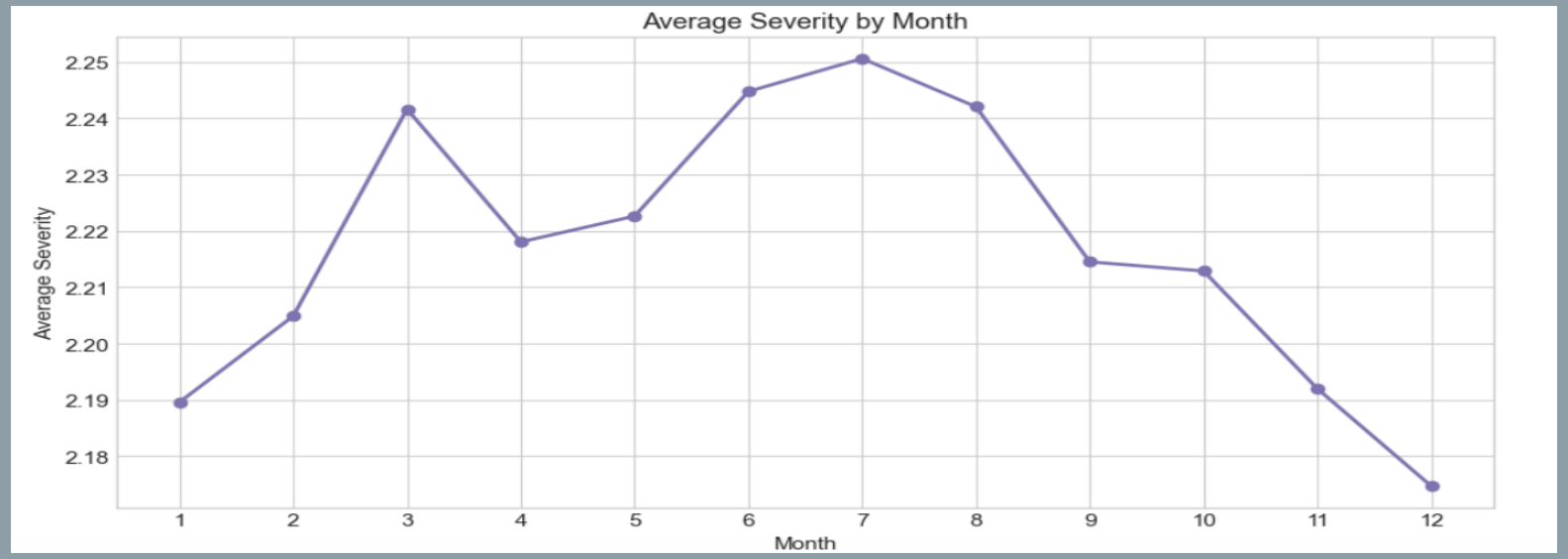




❑ National Comparison (from 200,000 samples)







Limits and Conclusion

- Results reflect conditional associations, not causal effects.
- Some factors such as driver behavior and speed are not observed.
- Only reported crashes are included.
- Texas results may differ from other states.

- Rain, wind speed, and junctions are linked to higher crash severity.
- Traffic signals and crossings are linked to lower severity.
- Results are consistent across models.
- Weather and road design matter for traffic safety.

THANK YOU!